

UNIRATIONAL NON-RATIONAL VARIETIES

[after M. ARTIN and D. MUMFORD]

by Pierre DELIGNE

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This exposé contains a description of the variety constructed by Artin and Mumford, and the proof that it is unirational and non-rational. It also contains the statements of the theorems proved by Clemens and Griffiths [4], and by Manin and Iskovskih [6], for constructing other examples. I attended talks by Artin and Mumford on the same subject and drew heavily on them.

In everything that follows, the expression “algebraic variety” will mean “a separated scheme of finite type over $\mathbb{C}$, reduced and irreducible”. We shall write $k(X)$ for the field of rational functions on an algebraic variety X .

1. Rationality and unirationality

Let K be an extension of finite type of $\mathbb{C}$. There then exists an algebraic variety X such that K is isomorphic to $k(X)$. X is called a *model* of K . By [5], K always admits a projective and smooth model.

Let K and L be two extensions of finite type of $\mathbb{C}$, with models X and Y . The homomorphisms $a^* : K \hookrightarrow L$ correspond bijectively to the dominant rational maps $a_* : Y \dashrightarrow X$, by the formula $a^*(f) = f \circ a_*$. By [5], if X and Y are projective and smooth, every rational map $a_* : Y \dashrightarrow X$ admits a decomposition

$$(1.1) \quad \begin{array}{ccccccc} Y_{n+1} & \xrightarrow{c_n} & \dots & \rightarrow & Y_2 & \xrightarrow{c_1} & Y_1 = Y \\ b \downarrow & & & & & & \\ & & & & & & X \end{array}$$

where b and the c_i are morphisms (defined everywhere), and where Y_{i+1} is obtained from Y_i by blowing up a smooth subvariety Z_i (of codimension ≥ 2).

If $a^* : K \rightarrow L$ is an isomorphism, the preceding result can be applied to $((a^*)^{-1}, Y_{n+1}, X)$. By iteration, one obtains a commutative diagram of morphisms of schemes

$$(1.2) \quad \begin{array}{ccccccc} \dots & Y^2 & \xrightarrow{y^2} & Y^1 & = & Y^1 & \xrightarrow{y^1} & Y \\ & \downarrow a^2 & & \uparrow b^1 & & \downarrow a^1 & & \downarrow a \\ \dots & X^1 & = & X^1 & \xrightarrow{x^1} & X & = & X \end{array}$$

In this diagram, the horizontal arrows x^i, y^i are composites of blow-ups with nonsingular centers, as in (1.1); the a^i and b^i are birational.

An invariant of a projective and smooth variety X is called *birational* if it depends only on $k(X)$. The birational character of an invariant is often verified with the aid of (1.2) (for a typical example, see 2.1). Invariants usable for a field K are generally defined as the value of a birational invariant of any projective and smooth model of K .

DEFINITION 1.3.- Let X be an algebraic variety of dimension n . We say that X is unirational if the following conditions are satisfied:

- (i) $k(X)$ is a subfield of a purely transcendental extension $\mathbb{C}(T_1, \dots, T_r)$ of $\mathbb{C}$;
- (ii) a finite extension of $k(X)$ is isomorphic to $\mathbb{C}(T_1, \dots, T_n)$;
- (iii) (for X projective and smooth), there exists a diagram (1.1) with $Y = \mathbb{P}^r(\mathbb{C})$;
- (iv) (for X projective and smooth), there exists a diagram (1.1) with $Y = \mathbb{P}^n(\mathbb{C})$.

One may suppose X projective and smooth, and one has

$$(ii) \Rightarrow (i) \Leftrightarrow (iii) \Rightarrow (iv) \Leftrightarrow (ii).$$

DEFINITION 1.4.- We say that X is rational if, alternatively,

- (i) $k(X)$ is a purely transcendental extension of $\mathbb{C}$;
- (ii) (for X projective and smooth), there exists a diagram (1.2) with $Y = \mathbb{P}^n(\mathbb{C})$.

The most evident birational invariants do not permit one to distinguish unirational varieties from rational varieties.

PROPOSITION 1.5.- Let X be a projective and smooth unirational algebraic variety of dimension $n > 0$.

- (i) The plurigenera

$$P_k = \dim H^0(X, (\Omega_X^n)^{\otimes k}) \quad (k > 0)$$

vanish. More generally, $H^0(X, (\Omega_X^1)^{\otimes k}) = 0$ for $k > 0$ (see [9], Remark, page 02).

- (ii) X is simply connected ([10]).

For $n = 1$ or 2 , property (i) above already characterizes rational varieties. A unirational variety of dimension ≤ 2 is therefore rational, and a subextension of transcendence degree ≤ 2 of $\mathbb{C}(T_1, \dots, T_k)$ is automatically purely transcendental. More precisely, one has the following:

- a) For $n = 1$, if $H^0(X, \Omega_X^1) = 0$, then $X \simeq \mathbb{P}^1(\mathbb{C})$. By the Riemann–Roch theorem, for every $x \in X$, $\mathcal{O}(x)$ indeed defines an embedding of X into $\mathbb{P}^1(\mathbb{C})$.
- b) For $n = 2$, the Castelnuovo–Enriques criterion asserts that X is rational if $P_a \simeq P_2 = 0$, i.e. if

$$H^0(X, \Omega_X^1) = H^0(X, (\Omega_X^2)^{\otimes 2}) = 0$$

(see [9]).

The symbols $P_a \simeq P_2$ and Ω_X^1 in this criterion reproduce the printed source.

For $n > 2$, it has been very difficult to define or calculate birational invariants capable of distinguishing between rational and unirational varieties.

2. Some birational invariants

A) Artin and Mumford

PROPOSITION 2.1.- The torsion subgroup $H^3(X, \mathbb{Z})_{\text{tors}}$ of $H^3(X, \mathbb{Z})$ is a birational invariant of the projective and smooth variety X .

If a variety X_2 is obtained from a smooth variety X_1 by blowing up a smooth subvariety $Z \subset X_1$ of codimension $r \geq 2$, then (SGA 5 VII)

$$H^n(X_2, \mathbb{Z}) \simeq H^n(X_1, \mathbb{Z}) \oplus \bigoplus_{i=1}^{r-1} H^{n-2i}(Z, \mathbb{Z}).$$

The lower summation index 1 , without an explicit $i =$, is retained from the printed formula. In particular,

$$H^3(X_2, \mathbb{Z}) \simeq H^3(X_1, \mathbb{Z}) \oplus H^1(Z, \mathbb{Z}).$$

Since $H^1(Z, \mathbb{Z})$ is torsion-free, one therefore has

$$H^3(X_1, \mathbb{Z})_{\text{tors}} \simeq H^3(X_2, \mathbb{Z})_{\text{tors}}.$$

Suppose that $k(X)$ is isomorphic to $k(Y)$, and let there be a diagram (1.2) From it one obtains a diagram

$$\begin{array}{ccccc}
H^3(X, \mathbb{Z})_{\text{tors}} & \xrightarrow{\sim} & H^3(X_1, \mathbb{Z})_{\text{tors}} & \xlongequal{\quad} & H^3(X_1, \mathbb{Z})_{\text{tors}} & \xrightarrow{\sim} & H^3(X_2, \mathbb{Z})_{\text{tors}} \\
& & \uparrow a^{1*} & & \downarrow b^{1*} & & \uparrow a^{2*} \\
H^3(Y, \mathbb{Z})_{\text{tors}} & = & H^3(Y, \mathbb{Z})_{\text{tors}} & \xrightarrow{\sim} & H^3(Y_1, \mathbb{Z})_{\text{tors}} & = & H^3(Y_1, \mathbb{Z})_{\text{tors}} .
\end{array}$$

Since $a_2^* b_1^*$ is bijective, b_1^* is injective. Since $b_1^* a_1^*$ is bijective and b_1^* injective, a_1^* is bijective, and the assertion follows.

The example of Artin and Mumford is based on the

CONSTRUCTION 2.2.- We shall construct a unirational variety X , projective and smooth, of dimension 3, such that

$$H^3(X, \mathbb{Z})_{\text{tors}} \neq 0 \quad (\text{we shall have } H^3(X, \mathbb{Z}) \simeq \mathbb{Z}/(2)).$$

Multiplying this variety by $\mathbb{P}^r(\mathbb{C})$, one obtains an analogous example in every dimension ≥ 3 .

B) Clemens and Griffiths

Let X be a projective and smooth variety of dimension 3, such that $H^0(X, \Omega_X^3) = 0$. The Hodge decomposition of $H^3(X, \mathbb{C})$ then reduces to

$$H^3(X, \mathbb{C}) = H^{2,1} \oplus H^{1,2}$$

and, according to Weil [11], the complex torus

$$J(X) = H^3(X, \mathbb{Z}) \backslash H^3(X, \mathbb{C}) / H^{2,1}$$

is an abelian variety: the *intermediate Jacobian* of X . Let

$$\psi : H^3(X, \mathbb{Z}) \otimes H^3(X, \mathbb{Z}) \longrightarrow H^6(X, \mathbb{Z}) \simeq \mathbb{Z}$$

be the cup product. By the Poincaré duality theorem, the alternating form on $H^3(X, \mathbb{Z})/\text{torsion}$ deduced from ψ has discriminant one. If $H^1(X, \mathbb{Z}) = 0$, it follows from Weil [11] that this form defines a principal polarization on $J(X)$.

PROPOSITION 2.3 ([4] 3.26).- If X is rational, the polarized abelian variety $J(X)$ is a product of Jacobians.

In the proof sketch that follows, the sign $\simeq$ denotes an isomorphism of *polarized* abelian varieties. One verifies successively:

a) $J(\mathbb{P}^3(\mathbb{C})) = 0$.

b) If Y'' is obtained from the nonsingular projective variety Y' by blowing up a smooth curve Z with Jacobian $J(Z)$ (respectively, a point), one has ([4] 3.11)

$$J(Y'') \simeq J(Y') \times J(Z) \quad (\text{respectively, } J(Y'') \simeq J(Y')).$$

c) If a morphism $b : Y' \rightarrow X$ is birational, $J(X)$ is a direct factor in $J(Y')$: there exists a principally polarized abelian variety of the principal series B such that

$$J(Y') \simeq J(X) \times B.$$

d) Let $(A_i)_{1 \leq i \leq n}$ be nonzero principally polarized abelian varieties. Suppose that no A_i admits a decomposition $A_i \simeq A'_i \times A''_i$ (for example, that A_i is a Jacobian). Then every decomposition of $A = \prod A_i$ is of the form

$$A \simeq \left(\prod_{i \in I} A_i \right) \times \left(\prod_{i \notin I} A_i \right) \quad \text{for } I \subset [1, n] \quad ([4] \text{ 3.23}).$$

If X is rational, there exists a diagram 2.1 with $Y = \mathbb{P}^3(\mathbb{C})$ and b birational. By a) and b), $J(Y_{n+1})$ is a product of Jacobians. By c) and d), $J(X)$ is one as well. The reference “diagram 2.1” reproduces the printed source.

Take X to be a cubic hypersurface in $\mathbb{P}^4(\mathbb{C})$. One has

$$H^1(X, \mathbb{Z}) = H^3(X, \mathbb{Z})_{\text{tors}} = H^0(X, \Omega_X^3) = 0, \quad \dim J(X) = \dim H^1(X, \Omega_X^2) = 5,$$

and X is unirational by [7]. Let S be the surface parametrizing the lines contained in X , and let $\alpha : S \rightarrow \text{Alb}(S)$ be the map of S into its Albanese variety. One of the essential results of [4] is the following

THEOREM 2.4 ([4] 11.19 AND 13.4).- There exists an isomorphism

$$J(X) \xrightarrow{\sim} \text{Alb}(S),$$

which carries the divisor Θ of $J(X)$ to the image of $S \times S$ under the map

$$(x, y) \xrightarrow{\delta} \alpha(x) - \alpha(y).$$

Clemens and Griffiths then manage to use results of [1] to prove that $(\text{Alb}(S), \delta(S \times S))$ cannot be the polarized Jacobian of a curve. Another proof of the irrationality of X is suggested in the appendix to [4]. It is a matter of proving

THEOREM 2.5.- The singular locus of the divisor Θ of $J(X)$ has dimension 0, hence codimension > 4 . It is even reduced to the image under δ of the diagonal of $S \times S$.

It is known that this is never the case for the divisor Θ of a Jacobian.

C) Manin and Iskovskih

The group of birational automorphisms of a variety X (equal to the group of $\mathbb{C}$ -automorphisms of $k(X)$) is a birational invariant of X . It is this invariant that Manin and Iskovski use to show that a smooth quartic hypersurface in $\mathbb{P}^4(\mathbb{C})$ is never rational:

THEOREM 2.6 ([6]).- Let X and Y be two smooth quartic hypersurfaces in $\mathbb{P}^4(\mathbb{C})$. Every birational isomorphism $a : X \rightarrow Y$ is automatically biregular.

Since certain quartic hypersurfaces are unirational, [7] this theorem provides a new example of unirational non-rational varieties.

3. The example of Artin and Mumford

Consider in $\mathbb{P}^2(\mathbb{C})$ a configuration consisting of

- a) two smooth cubic curves C_1 and C_2 , with equations $f_1 = 0$ and $f_2 = 0$, meeting transversely;
- b) a smooth conic Q , with equation $q = 0$, which meets each C_i in three distinct points of tangency.

We shall construct

- a) a vector bundle V of rank 3 on $\mathbb{P}^2(\mathbb{C})$;
- b) a quadratic form $\Phi : V \rightarrow \mathcal{L}$ ($\mathcal{L}$ an invertible sheaf) of discriminant $f_1 f_2$ (if Δ , a section of $\mathcal{L}^{\otimes 3} \otimes (\bigwedge^3 V)^{\otimes (-2)}$, is the discriminant, the subscheme of $\mathbb{P}^2(\mathbb{C})$ with equation $\Delta = 0$ is $C_1 \cup C_2$).

Let $X \subset \mathbb{P}(V^*)$ be the conic bundle over $\mathbb{P}^2(\mathbb{C})$, degenerating along $C_1 \cup C_2$, with homogeneous equation $\Phi = 0$. The following conditions will be satisfied.

- c) Φ has rank ≥ 2 at every point: for $s \in C_1 \cup C_2$, the fibre $X_s = f^{-1}(s)$ is the union of two concurrent lines; the only singular points of X are the singular points of the fibres X_s for $s \in C_1 \cap C_2$.
- d) Let S be the double cover of $\mathbb{P}^2(\mathbb{C})$, ramified along Q , and let $X_S = X \times_{\mathbb{P}^2(\mathbb{C})} S$. Then X_S/S has a section

$$\begin{array}{ccc} & & X \\ & \nearrow \alpha & \downarrow \\ S & \xrightarrow{\pi} & \mathbb{P}^2(\mathbb{C}) \end{array} .$$

- e) For $s \in (C_1 \cup C_2) - Q$, the two points $\alpha(t)$ for which $\pi(t) = s$ lie in the two irreducible components of X_s .

Locally (for the usual topology) on $C_1 \cup C_2$, the restriction of the bundle X to $C_1 \cup C_2$ is obtained by gluing two projective-line bundles along a section. However, when one traverses a loop in $C_1 \cup C_2$, these two bundles may be exchanged: there exists an unramified double cover $(C_1 \cup C_2)^*$ of $C_1 \cup C_2$ whose two local sections correspond, locally, to the two projective-line bundles of which $X|_{C_1 \cup C_2}$ is the union. Let $\pi_i : C_i^* \rightarrow C_i$

be the double cover of C_i induced by $(C_1 \cup C_2)^*$. We shall use from e) only the following corollary

LEMMA 3.1.- C_i^* is irreducible.

By e), C_i^* is the normalization of the double cover of C_i induced by S . The lemma follows from the fact that the section q of $\mathcal{O}(2)|_{C_i}$ is not the square of a section of $\mathcal{O}(1)|_{C_i}$; if q were a square, the three intersection points of Q with C_i would indeed be collinear.

To the double cover C_i^* of C_i correspond a character of order 2 of the fundamental group of C_i and a local system A_i , locally isomorphic to $\mathbb{Z}$, defined by the exact sequence of sheaves on C_i

$$0 \longrightarrow A_i \longrightarrow \pi_{i*} \mathbb{Z} \xrightarrow{\text{Tr}} \mathbb{Z} \longrightarrow 0,$$

where Tr is the “sum of the values on the two sheets of the covering”. One has

$$H^0(C_i, A_i) = 0, \quad H^1(C_i, A_i) = \mathbb{Z}/2, \quad H^2(C_i, A_i) = \mathbb{Z}/2. \quad (3.2)$$

We shall use d) only to prove the following fact

LEMMA 3.3.- X is unirational.

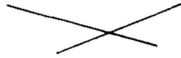
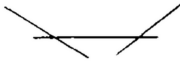
The surface S is a quadric in $\mathbb{P}^3(\mathbb{C})$, hence is rational. Since X_S/S has a section, there is a birational isomorphism $X_S \sim S \times \mathbb{P}^1(\mathbb{C})$, and X is the image of the rational variety X_S .

The nine singular points of X are ordinary quadratic points. Each can be resolved as follows (cf. [3])

- a) Blow it up; this resolves X and replaces the singular point by a quadric P , isomorphic to $\mathbb{P}^1(\mathbb{C}) \times \mathbb{P}^1(\mathbb{C})$.
- b) Choose a projection $P \xrightarrow{\beta} \mathbb{P}^1(\mathbb{C})$, and contract in the blow-up $\tilde{X}_1$ of X the fibres of this projection; this is possible because β is smooth, because its fibres are isomorphic to $\mathbb{P}^1(\mathbb{C})$, and because the normal bundle of P in $\tilde{X}_1$ induces on each fibre a bundle $\mathcal{O}(-1)$ ([8]).

Let $\tilde{X}$ be the resulting variety. There is a map from $\tilde{X}$ into X , and $\tilde{X}$ is obtained from X by replacing each singular point by a projective line.

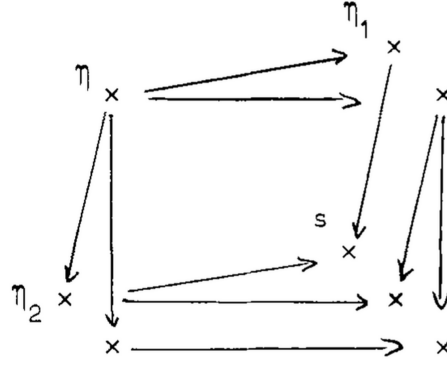
The fibres of the projection $p : \tilde{X} \rightarrow \mathbb{P}^2(\mathbb{C})$ are of the following three types

$s \notin C_1 \cup C_2$		une conique
$s \in C_1 \cup C_2$, $s \notin C_1 \cap C_2$		deux droites concourantes
$s \in C_1 \cap C_2$		trois droites comme indiqu��.

English key: $s \notin C_1 \cup C_2$, a conic; $s \in C_1 \cup C_2$ and $s \notin C_1 \cap C_2$, two concurrent lines; $s \in C_1 \cap C_2$, three lines as shown.

When one passes from the generic point η of $\mathbb{P}^2(\mathbb{C})$ to the generic point η_i of C_i ($i = 1, 2$), and to $s \in C_1 \cap C_2$, the irreducible components specialize as indicated in the following diagram

(3.4)



(the crosses represent irreducible components, all the multiplicities being one). The tedious verification of this local result is omitted. From it one obtains

LEMMA 3.5.- One has

$$R^0 p_* \mathbb{Z} = \mathbb{Z}, \quad R^1 p_* \mathbb{Z} = 0,$$

and an exact sequence of sheaves

$$0 \longrightarrow A_1 \oplus A_2 \longrightarrow R^2 p_* \mathbb{Z} \xrightarrow{\text{Tr}} \mathbb{Z} \longrightarrow 0.$$

We can now use the Leray spectral sequence of p to compute $H^*(\tilde{X}, \mathbb{Z})$. The essential fact is the following (which implies the irrationality of X).

LEMMA 3.6.- One has

$$H^3(\tilde{X}, \mathbb{Z}) = \mathbb{Z}/(2).$$

Let us compute the E_2 -terms of the Leray spectral sequence of p . One has

$$\begin{aligned} R^0 p_* \mathbb{Z} = \mathbb{Z} \quad \text{hence} \quad E_2^{p,0} &= \mathbb{Z}, 0, \mathbb{Z}, 0, \mathbb{Z} \quad \text{for } p = 0 \text{ to } 4, \\ R^1 p_* \mathbb{Z} = 0 \quad \text{hence} \quad E_2^{p,1} &= 0. \end{aligned}$$

To compute $E_2^{p,2} = H^p(R^2 p_* \mathbb{Z})$, one uses the long exact sequence defined by the short exact sequence of sheaves 3.5. One finds

$$\begin{aligned} 0 \longrightarrow E^{0,2} \xrightarrow{a} \mathbb{Z} \xrightarrow{j} \mathbb{Z}/2 \oplus \mathbb{Z}/2 \longrightarrow E^{1,2} \longrightarrow 0, \\ 0 \longrightarrow \mathbb{Z}/2 \oplus \mathbb{Z}/2 \longrightarrow E^{2,2} \longrightarrow \mathbb{Z} \longrightarrow 0, \\ E^{3,2} = 0 \quad \text{and} \quad E^{4,2} = \mathbb{Z}. \end{aligned}$$

One easily verifies that $a(E^{0,2}) = 2\mathbb{Z}$ (this fact is, moreover, not necessary for the proof that $H^3(\tilde{X}, \mathbb{Z})_{\text{tors}} \neq 0$). The E_2^{pq} are therefore as follows

$q \uparrow$	$2\mathbb{Z}$	$\mathbb{Z}/2$	$E^{2,2}$	0	$\mathbb{Z}$
0	0	0	0	0	0
$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	$\mathbb{Z}$
	$p \rightarrow$				

The differentials d_r can only be zero, so that $E_2^{pq} = E_\infty^{pq}$, and the lemma follows.

It is apparent that it was essential to have two curves $C_i \subset \mathbb{P}^2(\mathbb{C})$ along which X degenerates, for a group $\mathbb{Z}/2$ is killed in passing from

$$H^1(\text{Ker}(R^2 p_* \mathbb{Z} \rightarrow \mathbb{Z})) \quad \text{to} \quad H^1 R^2 p_* \mathbb{Z} = E^{1,2}.$$

It remains to carry out the promised constructions. Artin and Mumford know how to prove *a priori* the existence of a conic bundle of the required type. I shall confine myself to giving equations that define one.

- 1) The image of $f_1 f_2$ in $H^0(Q, \mathcal{O}(6))$ is the square of $g_1 \in H^0(Q, \mathcal{O}(3))$, because Q is rational and the zeros of $f_1 f_2$ on Q are double. This g_1 lifts to $g \in H^0(\mathbb{P}^2(\mathbb{C}), \mathcal{O}(3))$ because $H^1(\mathbb{P}^2(\mathbb{C}), \mathcal{O}(1)) = 0$. Since $f_1 f_2 - g^2$ vanishes on Q , it is a multiple of q :

$$f_1 f_2 = g^2 + qd.$$

- 2) Take

$$V = \mathcal{O}(-1) \oplus \mathcal{O}(-2) \oplus \mathcal{O},$$

and the form

$$\Phi : V \longrightarrow \mathcal{O} : (x, y, z) \longmapsto qx^2 + 2gxy - dy^2 - z^2.$$

One has $\Delta = g^2 + qd = f_1 f_2$. At a point where Φ had rank ≤ 1 , one would have $\Delta = q = d = 0$, whence $g = 0$, and $\Delta = g^2 + qd$ would have a double zero. This is impossible because $C_1 \cap C_2 \cap Q = \emptyset$. Finally, S is identified with the subscheme of X having equation $y = 0$, and d) and e) follow.

BIBLIOGRAPHY

- [1] A. Andreotti, On a theorem of Torelli, *Am. J. of Math.*, 80 4 (1958), p. 801–828.
- [2] M. Artin and D. Mumford, Some elementary examples of unirational varieties which are not rational, *Journal London Math. Soc.*, (to appear).
- [3] M. Atiyah, On analytic surfaces with double points, *Proc. Roy. Soc. London*, Ser. A 247 (1958), p. 237–244.
- [4] C. H. Clemens and P. A. Griffiths, The intermediate jacobian of the cubic threefold, preprint.
- [5] H. Hironaka, Resolution of singularities of an algebraic variety over a field of characteristic zero: I, II, *Ann. of Math.*, 79 (1964), p. 109–326.
- [6] Ju. I. Manin et V. A. Iskovskih, L’hypersurface quartique de dimension trois, et un contre-exemple au problème de Lüroth, *Mat. Sbornik*, 86 1 (1971), 140–166.
- [7] L. Roth, *Algebraic threefolds*, Ergebnisse der Math., Heft 6, Springer-Verlag, 1955.
- [8] B. Saint-Donat, Sur un théorème de G. Castelnuovo et F. Enriques, Thèse de 3ème Cycle, Lyon 1968.
- [9] J.-P. Serre, Critère de rationalité pour les surfaces algébriques (d’après K. Kodaira), Séminaire Bourbaki, exposé 146, volume 1956/57, W. A. Benjamin, New York.
- [10] J.-P. Serre, On the fundamental group of a unirational variety, *J. London Math. Soc.* 34 (1959), p. 481–484.
- [11] A. Weil, *Introduction à l’étude des variétés kählériennes*, Hermann, Paris 1958.